Detecting Suspicious Reviews/ Potentially Fake Reviews

Course: Data Management for Data Science
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1 Project Definition

1.1 Problem Statement

Online review platforms like Yelp play an important role in how users evaluate businesses. People rely heavily on ratings and written reviews when making decisions, which makes the reliability of this information critical.However, not all reviews reflect genuine experiences. Some reviews may be misleading, biased, or inconsistent with the overall perception of a business. These reviews can distort user trust and influence decision-making in unfair ways.A key challenge is that such reviews are not always obvious and may only become noticeable when looking at patterns across user behavior, ratings, and review characteristics.

1.2 Connection to Course Material

This project directly connects to several concepts we learned in class, especially around data management and building a complete data pipeline. We started by working with raw Yelp data, which required us to apply data cleaning techniques such as filtering, handling missing values, and structuring the data in a usable format. This step closely reflects the data preprocessing methods we practiced in class.We also worked with multiple datasets, including review, user, and business data, and combined them using common identifiers. This relates to relational data concepts and schema design, where different entities are connected to build a more meaningful dataset. To further strengthen this, we stored our processed data in a SQLite database and used SQL queries to explore patterns such as user activity and business behavior.In addition, we applied feature engineering by creating new variables such as review length, rating deviation, and user activity. This reflects the idea that raw data often needs to be transformed before it can be used effectively for analysis or modeling.From a modeling perspective, we applied concepts related to unsupervised learning and anomaly detection. Since the dataset did not contain labels, we used Isolation Forest to identify unusual patterns instead of predicting fixed outputs. Overall, this project helped us connect different parts of the course, from data preparation and SQL to feature engineering and machine learning, into one complete workflow.

2 Novelty and Importance

2.1 Importance of the Project

Online review platforms like Yelp play a major role in how users make decisions about businesses. Research and industry discussions around online platforms have consistently highlighted the impact of fake or misleading reviews, especially because users tend to rely more on extreme reviews when forming opinions.Even a small number of suspicious reviews can influence perception disproportionately. For example, highly positive or highly negative reviews often stand out more than neutral ones, which means they can affect user trust and decision-making even if they do not significantly change the overall rating of a business.Because of this, detecting suspicious review behavior is important not just for maintaining data quality, but also for ensuring fairness and transparency on online platforms.

The distribution shows a clear skew toward higher ratings, with most reviews being 4 or 5 stars. This imbalance suggests that even a small number of low or extreme ratings can stand out and disproportionately influence user perception.

2.2 Prior Work and Gap

Previous approaches to fake review detection often relied on supervised learning methods, where models are trained using labeled datasets that clearly indicate whether a review is fake or genuine. These approaches typically use advanced natural language processing techniques and classification models.However, in real-world scenarios, labeled datasets are not always available, and creating such labels can be time-consuming and unreliable. Many fake reviews are also designed to look realistic, which makes purely text-based detection difficult.In our project, we address this gap by focusing on **unsupervised anomaly detection**, where the goal is to identify unusual patterns without relying on labels. Instead of trying to directly classify reviews as fake or real, we analyze behavioral signals such as rating deviation, user activity, and review structure.This approach is more practical for real-world applications, where the goal is often to flag suspicious patterns for further investigation rather than make definitive classifications.

2.3 What Makes Our Approach Different

Our approach goes beyond just analyzing review text by incorporating user behavior, business context, and review patterns. By combining multiple data sources and using SQL for exploration, we build a more complete understanding of the data. This allowed us to identify suspicious patterns more effectively rather than relying on a single dimension.

3 Progress and Contribution

3.1 Data Utilization and Column Selection

We used an open dataset from Yelp which is a subset of Yelp data that is intended for educational use. It provides real-world data related to businesses including reviews, photos, check-ins, and attributes like hours, parking availability, and ambience.The Yelp dataset consisted of multiple JSON files. For this particular project, we focused on the review, business, and user datasets, as they contain the most relevant information for our model. While additional datasets such as tip and checkin were available, they were not used in this project because they do not directly contribute to detecting suspicious review patterns. Since processing the full dataset was not feasible as it was approximately 9GB large, with around 6,990,280 reviews, we sampled approximately 20,000 reviews.We carefully selected only the most relevant columns instead of using everything. This decision was based on our understanding of how suspicious behavior might appear.We included review text and star ratings to capture user feedback business average rating and review count to understand expected behavior user review count and average rating to capture user patterns. We chose these features because suspicious reviews often appear as inconsistencies. For example, a review that strongly deviates from a business's average rating may indicate abnormal behavior.

REVIEW DATASET SAMPLE DATA:

	review_id	user_id	business_id	stars	useful	funny	cool	text	date
0	KU_O5udG6zpxOg-VcAEodg	mh_-eMZ6K5RLWhZyISBhwA	XQfwVwDr-v0ZS3_CbbE5Xw	3.0	0	0	0	If you decide to eat here, just be aware it is...	2018-07-07 22:09:11
1	BiTunyQ73aT9WBnpR9DZGw	OyoGAe7OKpv6SyGZT5g77Q	7ATYjTIgM3jUlt4UM3IypQ	5.0	1	0	1	I've taken a lot of spin classes over the year...	2012-01-03 15:28:18

Figure 1: Review Dataset

BUSINESS DATASET SAMPLE DATA:

	business_id	name	address	city	state	postal_code	latitude	longitude	stars	review_count	is_open	attributes	categories	hours
0	Pns2l4eNsfO8kk83dixA6A	Abby Rappoport, LAC, CMQ	1616 Chapala St, Ste 2	Santa Barbara	CA	93101	34.426679	-119.711197	5.0	7	0	{'ByAppointmentOnly': 'True'}	Doctors, Traditional Chinese Medicine, Naturop...	None
1	mpf3x-BjTdTEA3yCZrAYPw	The UPS Store	87 Grasso Plaza Shopping Center	Affton	MO	63123	38.551126	-90.335695	3.0	15	1	{'BusinessAcceptsCreditCards': 'True'}	Shipping Centers, Local Services, Notaries, Ma...	{'Monday': '0:0-0:0', 'Tuesday': '8:0-18:30', ...

Figure 2: Business Dataset

```
USER DATASET COLUMNS:

['user_id', 'name', 'review_count', 'yelping_since', 'useful', 'funny', 'cool', 'elite', 'friends', 'fans', 'average_stars', 'compliment_hot', 'compliment_more', 'compliment_profil
e', 'compliment_cute', 'compliment_list', 'compliment_note', 'compliment_plain', 'compliment_cool', 'compliment_funny', 'compliment_writer', 'compliment_photos']

USER DATASET SAMPLE DATA:
```

	user_id	name	review_count	yelping_since	useful	funny	cool	elite	friends	fans	...	compliment_more	compliment_profile
0	qVc8ODYU5SZjKXVBgXdI7w	Walker	585	2007-01-25 16:47:26	7217	1259	5994	2007	NSCy54eWehBJyZdG2iE84w, pe42u7DcCH2QmI81NX-8qA...	267	...	65	55
1	j14WgRoU_-2ZE1aw1dXrJg	Daniel	4333	2009-01-25 04:35:42	43091	13066	27281	2009,2010,2011,2012,2013,2014,2015,2016,2017,2...	ueRPE0CX75ePGMqOFVj6IQ, 52oH4DrRvzzl8wh5UXyU0A...	3138	...	264	184

2 rows x 22 columns

Figure 3: User Dataset

```
['user_id', 'business_id', 'text', 'date', 'compliment_count']

TIP DATASET SAMPLE DATA:
```

	user_id	business_id	text	date	compliment_count
0	AGNUgVwnZUey3gcPCJ76iw	3uLgwr0qeCNMjKenHJwPGQ	Avengers time with the ladies.	2012-05-18 02:17:21	0
1	NBN4MgHP9D3cw--SnauTkA	QoezRbYQncpRqyrLH6Iqjg	They have lots of good deserts and tasty cuban...	2013-02-05 18:35:10	0

```
CHECKIN DATASET COLUMNS:

['business_id', 'date']

CHECKIN DATASET SAMPLE DATA:
```

	business_id	date
0	---kPU91CF4Lq2-WlRu9Lw	2020-03-13 21:10:56, 2020-06-02 22:18:06, 2020...
1	--0iUa4sNDFiZFrAdIWhZQ	2010-09-13 21:43:09, 2011-05-04 23:08:15, 2011...

Figure 4: Tip and Checkin Dataset(not used)

3.2 Data Cleaning and Preprocessing

As shown in *Figure 1*, we used the review dataset as the primary source since it contains the core information such as review text, ratings, and timestamps. However, analyzing reviews alone does not provide enough context, so we merged this data with the business(*Figure 2*) and user datasets(*Figure 3*). The business dataset helped us understand expected behavior through attributes like average ratings and review counts, while the user dataset allowed us to analyze user activity and patterns.We performed the merge using common identifiers, specifically business_id and user_id. The review dataset was treated as the main table, and we used left joins to ensure that all reviews were preserved even if some user or business information was missing. This was important because removing reviews due to missing values could result in loss of potentially useful patterns.After merging, we carefully selected only the columns that were relevant to our analysis. From the review dataset, we kept attributes such as review_id, user_id, business_id, stars, text, and date. We intentionally removed columns such as useful, funny, and cool because they represent user engagement rather than characteristics of the review itself, and therefore did not directly contribute to identifying suspicious behavior.From the business dataset, we selected attributes such as business average rating, review count, location, and categories, as these provide context for what is considered normal behavior for a business. From the user dataset, we kept features such as user review count and average stars, which help capture

user-level behavior. We also decided not to use the tip and check-in datasets. The tip dataset consists of short comments that are not structured like full reviews, and therefore do not provide meaningful signals for our analysis. The check-in dataset mainly contains timestamps of visits, which are not directly related to review behavior. Since our focus was on detecting suspicious reviews, these datasets were not relevant to our problem.After merging the datasets, we observed that some fields contained missing values, particularly in user-related attributes. This occurred because not all user_ids present in the review dataset had corresponding entries in the user dataset. Instead of removing these rows, we handled missing values carefully to preserve the dataset and avoid losing potentially useful patterns. For user-related features such as user_review_count, missing values were replaced with zero, allowing us to treat these users as having low or no activity. For other attributes, simple imputation methods such as replacing missing values with overall averages were used where appropriate. This ensured that the dataset remained consistent and usable for further analysis.Another major challenge was the size of the dataset. Since the full Yelp dataset is very large, we were not able to load it entirely into memory. To address this, we created a sampled dataset by reading the JSON file line by line and selecting the first 20,000 reviews. This allowed us to work with a manageable subset while still preserving meaningful patterns.This processed and merged dataset was saved as a CSV file referred to as sampled_reviews.csv. This file contains the cleaned and structured data after merging and preprocessing, and serves as the base dataset for feature engineering and model development.

```
Sampled Dataset Columns:

['review_id', 'user_id', 'business_id', 'stars', 'text', 'date', 'name', 'business_avg_stars', 'business_review_count', 'categories', 'city', 'state', 'user_name', 'user_review_count', 'average_stars',
'yelping_since', 'fans']
```

review_id	user_id	business_id	stars	text	date	name	business_avg_stars	business_review_count	categories	city	state	user_name	user_review_count	average_stars	yelping_since	fans
g-VcAEodg	mh_-eMZ6K5RLWhZyISBhwA	XQfwVwDr-v0ZS3_CbbE5Xw	3.0	If you decide to eat here, just be aware it is...	2018-07-07 22:09:11	Turning Point of North Wales	3.0	169	Restaurants, Breakfast & Brunch, Food, Juice B...	North Wales	PA	NaN	NaN	NaN	NaN	NaN
pR9DZGw	OyoGAe7OKpv6SyGZT5g77Q	7ATYjTIgM3jUlt4UM3IypQ	5.0	I've taken a lot of spin classes over the year...	2012-01-03 15:28:18	Body Cycle Spinning Studio	5.0	144	Active Life, Cycling Classes, Trainers, Gyms, ...	Philadelphia	PA	Erin	10.0	4.3	2011-03-07 19:45:15	0.0

Figure 1: sampled_reviews.csv Columns

After feature engineering and model application, which is discussed in depth in later parts of this report we generated a second dataset called modeled_reviews.csv. This dataset includes all the original features along with additional engineered features such as review length, sentiment score, rating deviation, and the final anomaly label indicating whether a review was flagged as suspicious. This version of the dataset was used for visualization and analysis in the later stages of the project.

```
Modeled Dataset Columns:

['review_id', 'user_id', 'business_id', 'stars', 'text', 'date', 'name', 'business_avg_stars', 'business_review_count', 'categories', 'city', 'state', 'user_name', 'user_review_count', 'average_stars', 'yelping_since', 'fans', 'review_length', 'sentiment', 'rating_deviation_from_business', 'rati
ng_deviation_from_user', 'is_extreme_rating', 'kmeans_cluster', 'hierarchical_cluster', 'anomaly_flag', 'anomaly_score', 'is_suspicious']
```

review_id	user_id	business_id	stars	text	date	name	business_avg_stars	business_review_count	categories	...	review_length	sentiment	rating_deviation_from_business	rating_deviation_from_user	is_extreme_rating	kmeans_cluster	hierarchical_cluster	anomaly_flag	anomaly_score	is_suspicious
xOg-VcAEodg	mh_-eMZ6K5RLWhZyISBhwA	XQfwVwDr-v0ZS3_CbbE5Xw	3.0	If you decide to eat here, just be aware it is...	2018-07-07 22:09:11	Turning Point of North Wales	3.0	169	Restaurants, Breakfast & Brunch, Food, Juice B...	...	101	0	0.0	0.85235	0	0	0	1	0.142775	0
VBnpR9DZGw	OyoGAe7OKpv6SyGZT5g77Q	7ATYjTIgM3jUlt4UM3IypQ	5.0	I've taken a lot of spin classes over the year...	2012-01-03 15:28:18	Body Cycle Spinning Studio	5.0	144	Active Life, Cycling Classes, Trainers, Gyms, ...	...	151	2	0.0	0.70000	1	1	2	1	0.125541	0

Figure 2: modeled_reviews Coulmns

Feature Engineering
Feature engineering was a crucial step in our project, as the raw dataset alone did not capture the patterns needed to identify suspicious behavior. We created several derived features that represent both textual characteristics of reviews and behavioral patterns of users and businesses. One of the key features was review length, which helped us identify reviews that were unusually short or excessively long compared to the general distribution. We also included sentiment analysis to capture the emotional tone of the review text which allowed us to detect inconsistencies between the sentiment expressed in the text and the star rating given.In addition, we engineered features such as rating deviation, which measures how much a review's rating differs from the average rating of the business, and user activity, which reflects how frequently a user posts reviews. These features were particularly useful in capturing unusual patterns, such as extreme ratings or highly active users. Overall, feature engineering allowed us to transform the dataset into a more meaningful representation, making it easier for the model to detect anomalies based on multiple dimensions rather than relying on a single attribute.

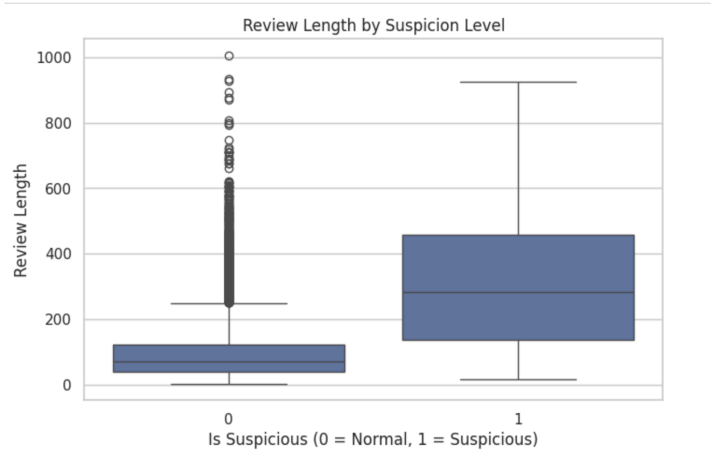
3.3 Models, Techniques, and Algorithms

K-Means Clustering We first applied K-Means clustering to group reviews based on similarity across the engineered features. The goal was to see if suspicious reviews would naturally form separate clusters from normal ones.While K-Means was effective in grouping reviews into clusters, it did not perform well for anomaly detection. Most data points were assigned to clusters regardless of how unusual they were, and suspicious reviews were not clearly isolated. This is mainly because K-Means assumes that clusters are evenly distributed and of similar density, which does not align with our dataset where anomalies are rare and scattered.As a result, K-Means grouped anomalous reviews with normal ones instead of separating them, making it difficult to identify suspicious behavior.
Hierarchical Clustering We then experimented with hierarchical clustering to explore whether a more flexible clustering structure would improve the results. This method allowed us to visualize nested groupings and relationships between reviews at different levels of similarity.However, the results were still not effective for identifying suspicious reviews. While hierarchical clustering provided better visualization of the data structure, the clusters were difficult to interpret, and there was no clear boundary separating normal and anomalous reviews.This approach also became computationally expensive and less practical as the dataset size increased. Overall, it did not provide a reliable way to isolate suspicious patterns.
Isolation Forest (Final Model) After observing the limitations of clustering methods, we shifted to Isolation Forest, which is specifically designed for anomaly detection.Unlike clustering algorithms, Isolation Forest does not attempt to group similar data points. Instead, it works by isolating observations that behave differently from the majority. Data points that are easier to isolate are considered anomalies.This approach was better suited for our dataset because suspicious reviews are expected to be rare and not form clear clusters. Our engineered features, such as rating deviation, review length, sentiment, and user activity, helped the model identify these unusual patterns.We set the contamination parameter to 2 percent, based on the assumption that only a small fraction of reviews are likely to be suspicious. This allowed the model to focus on the most extreme outliers.Isolation Forest performed significantly better than the clustering approaches because it directly targeted anomalies rather than trying to fit all data into structured

groups. It was also more consistent with our goal of identifying unusual behavior across multiple features rather than relying on similarity alone.

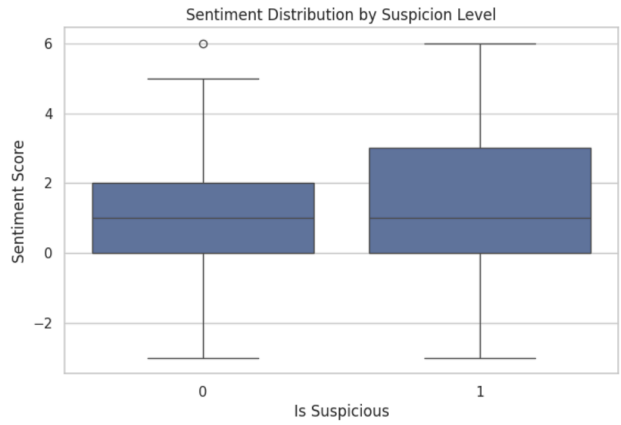
3.5 Results and Analysis

3.5.1 Review Length Analysis



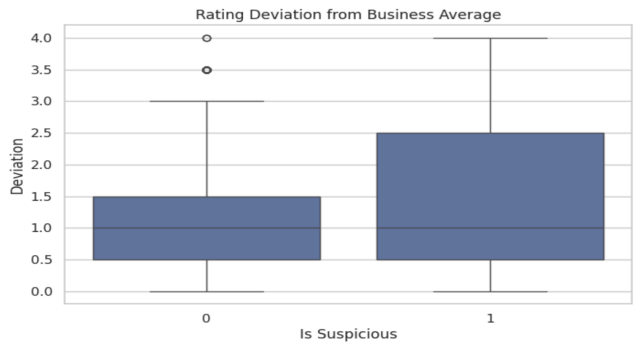
The analysis of review length revealed that suspicious reviews tend to exhibit a wider spread compared to normal reviews. While normal reviews were more concentrated within a moderate range of lengths, suspicious reviews showed both extremes, with some being very short and others being significantly longer. We observed that many flagged reviews are unusually long. This indicated that such reviews are either overly detailed or intentionally structured to appear more convincing. In some cases, longer reviews included excessive descriptions or repeated points, which could be a sign of unnatural writing behavior.However, it is important to note that review length alone is not a definitive indicator of suspicious activity. Many genuine users also write detailed reviews. The key observation here was that suspicious reviews tend to fall outside the typical range of lengths observed in the dataset.This suggested that review length is most useful when combined with other features, rather than being used independently.

3.5.2 Sentiment Analysis



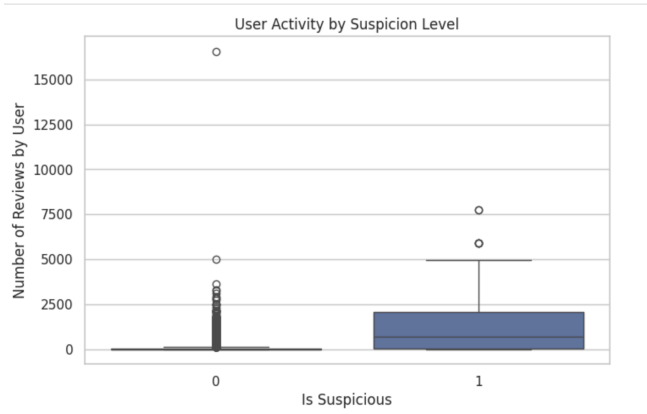
The sentiment distribution showed that both normal and suspicious reviews have overlapping sentiment ranges, indicating that sentiment alone is not a strong standalone indicator of

suspicious behavior. While suspicious reviews exhibited a slightly wider spread, including more extreme positive and negative values, the median sentiment was relatively similar across both groups. This suggested that suspicious reviews were not necessarily defined by their emotional tone, but rather by inconsistencies when sentiment combined with other features such as rating deviation and review length.



Rating deviation emerged as the strongest and most consistent indicator of suspicious behavior in our analysis. This feature measured how much a given review's rating differs from the average rating of the corresponding business.We observed that reviews flagged as suspicious tend to have significantly higher deviation compared to normal reviews. This indicated that these reviews do not align with the general consensus about a business.For example, a 1-star review for a business that consistently receives 4.5-star ratings, or a 5-star review for a poorly rated business, represents a strong deviation from expected behavior. Such inconsistencies indicated biased or manipulated reviews.This result strongly supported our decision to include rating deviation as a key feature. It also explained why the Isolation Forest model performed well, since it is designed to detect points that differ significantly from the majority.Overall, rating deviation provides a clear and interpretable signal of anomalous behavior.

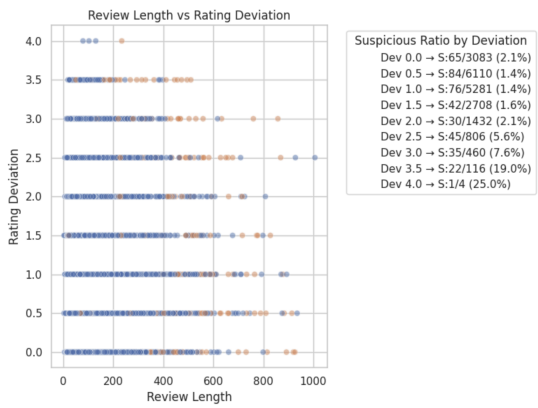
3.5.3 User Activity Analysis



User activity, measured by the number of reviews written by a user, also showed a noticeable pattern. We observed that suspicious reviews are more likely to be associated with users who have higher levels of activity.This suggested that certain users contribute disproportionately to

the dataset. In some cases, this may indicate coordinated behavior or users who are consistently posting reviews at a higher frequency than normal.However, similar to review length, this feature must be interpreted carefully. High activity does not necessarily imply suspicious intent, as some users may simply be more engaged or active on the platform.The key insight is that when high user activity is combined with other unusual patterns, such as high rating deviation or inconsistent sentiment, it becomes a stronger indicator of potential anomalies.

3.5.4 Combined Feature Interaction



The most important insight from our analysis came from examining how multiple features interact with each other.The scatter plot showed relationship between review length and rating deviation, with suspicious reviews highlighted. At low levels of rating deviation, most reviews appeared normal regardless of their length. This indicated when a review aligns with expected ratings, its length does not significantly affect its classification.However, as rating deviation increased, the proportion of suspicious reviews increased significantly. Within this high-deviation range, we also observed that many flagged reviews tend to have longer lengths.This suggested that suspicious behavior was not driven by a single feature, but rather by the combination of multiple unusual patterns. A review that was both highly inconsistent in rating and atypical in length is far more likely to be flagged as suspicious.This finding reinforced the importance of feature engineering in our project. Instead of relying on a single signal, our model benefited from combining behavioral and textual features to capture complex patterns in the data.

3.6 Key Findings and Insights

From the overall analysis, we found that suspicious reviews were not defined by extreme values in a single feature, but rather by inconsistencies across multiple dimensions of the data. Rating deviation emerged as the strongest indicator, showing that reviews that did not align with expected business ratings were more likely to be flagged. This highlighted the importance of understanding data in context rather than in isolation.We also observed that features such as review length, sentiment, and user activity were more effective when combined, even though they were not strong indicators individually. This indicated that suspicious behavior is better captured through interactions between features, rather than relying on any single attribute. From a broader perspective, these findings reinforced key concepts from the course, particularly the importance of data preprocessing, feature engineering, and multi-dimensional analysis. The results showed that raw data alone is often insufficient, and meaningful insights require

transforming and combining different aspects of the data. Additionally, the use of anomaly detection demonstrated how unsupervised methods can be applied in situations where clear labels are not available.Overall, this analysis emphasized that detecting suspicious patterns in real-world data requires both structured data management and thoughtful feature design, aligning closely with the concepts covered throughout the course.

3.7 Advantages and Limitations

One of the main **advantages** of our approach was the ability to detect suspicious patterns without relying on labeled data. By using anomaly detection, we were able to identify reviews that behaved differently from the majority, which is more realistic in real-world scenarios where ground truth labels are often unavailable. The combination of multiple features such as rating deviation, review length, sentiment, and user activity also allowed us to capture patterns across different dimensions of the data, rather than relying on a single indicator.Another advantage was the integration of multiple datasets, including review, business, and user data. This provided a more complete view of the system and enabled us to analyze behavior in context, which would not have been possible using a single dataset. Additionally, storing and querying the data using SQL helped improve our ability to explore patterns efficiently and reinforced structured data management practices.

However, there are several **limitations** to our approach. Since we used an unsupervised anomaly detection model, the results do not confirm whether a review is truly fake or genuine. Instead, the model identifies reviews that are unusual, which means that some flagged reviews may be legitimate but uncommon. This introduces the possibility of false positives.Another limitation is related to the features used in the model. While features such as review length and user activity helped capture patterns, they are not inherently indicators of suspicious behavior. For example, a genuine user may write detailed reviews or be highly active on the platform, which makes it difficult to distinguish between normal and abnormal behavior in some cases.The sentiment analysis component also has limitations, as it was implemented using a simple keyword-based approach due to environment constraints. This method may not fully capture complex language patterns such as sarcasm or context.Finally, the use of a sampled dataset instead of the full Yelp dataset may limit the generalizability of our findings. While sampling allowed us to work within computational constraints, it may not fully represent the diversity of the entire dataset.

Overall, while the model provides useful insights into suspicious patterns, it should be viewed as a tool for identifying anomalies rather than making definitive classifications.

4 Changes After Proposal

4.1 Differences from Proposal

After submitting our initial proposal, our approach evolved significantly based on both practical challenges and the feedback we received from our TAs, Naina and Chetan. In the proposal, we planned to work with the full Yelp dataset and use more advanced NLP-based techniques for analyzing review text. However, during implementation, we quickly realized that working with the complete dataset was not feasible due to its size, as it repeatedly caused memory issues and system crashes. Based on TA feedback, we shifted to using CodeBench and Rutgers iLab, which allowed us to handle larger files more efficiently. We also introduced a sampling strategy, where we limited our dataset to 20,000 reviews. This change made the project computationally manageable while still preserving meaningful patterns in the data.

Our feature engineering approach also became more focused after the proposal stage. Initially, we planned to rely more heavily on textual analysis, but after further discussion and feedback, we expanded our focus to include behavioral features such as rating deviation and user activity. This shift helped us better capture suspicious patterns beyond just the review text.In terms of modeling, our original plan included clustering methods as the primary approach. However, after experimenting with K-Means and hierarchical clustering, we found that these methods were not effective in isolating rare and scattered anomalies. With guidance from our TAs, we reevaluated our approach and transitioned to using Isolation Forest, which is better suited for anomaly detection.We also modified our sentiment analysis approach. Instead of using external NLP libraries as initially planned, we implemented a simpler keyword-based method due to environment limitations. While this was a simplification, it ensured that our pipeline remained functional and reproducible. Overall, the project shifted from an idealized plan to a more practical and well-structured implementation. The feedback from our TAs played a key role in helping us make these adjustments, allowing us to refine our approach and build a more realistic and effective solution.

4.2 Challenges and Bottlenecks

Throughout the project, we encountered several challenges that required us to rethink and refine our approach. One of the biggest challenges was handling the size and structure of the Yelp dataset. Loading and processing large JSON files was computationally expensive, and we had to redesign our workflow to read the data in smaller portions and filter only the necessary records.Another challenge was understanding the dataset itself. Since the data was spread across multiple files, we had to carefully analyze each dataset and decide which features were relevant for our problem. This required going through the data in detail and making decisions about what to include and what to exclude.We also faced challenges in model selection. The initial use of clustering techniques did not produce meaningful results for anomaly detection, which led us to experiment with different approaches before settling on Isolation Forest. This process involved trial and error and required us to evaluate the strengths and limitations of each method.Additionally, environment limitations, such as difficulty installing certain libraries, affected how we implemented sentiment analysis and forced us to adapt our approach. Overall, these challenges helped us refine our pipeline and made our final solution more practical and aligned with real-world constraints.

5 Team Contributions

From start to finish, we worked closely as a team and made sure to stay aligned throughout the project. We regularly communicated, often calling each other to divide tasks and discuss progress, which helped us stay organized and avoid overlap in our work. In the beginning, we worked together on understanding the dataset and setting up the preprocessing pipeline, since this was one of the most critical parts of the project. As we moved forward, we split responsibilities based on different stages of the pipeline. **Tamanna** focused more on data preprocessing, feature engineering, and structuring the datasets, while **Krishna** worked on implementing the models and refining the anomaly detection approach. Even though we had defined roles, we continuously reviewed each other’s work and made changes collaboratively to ensure consistency.

When we faced challenges, such as handling large datasets or deciding on the final model, we worked together to troubleshoot and explore different approaches. For example, while experimenting with clustering techniques and later transitioning to Isolation Forest, we both contributed to testing, evaluating, and understanding the results.Overall, our workflow involved both independent contributions and collaborative problem-solving. This helped us maintain steady progress and ensured that both of us had a clear understanding of all parts of the project.

6 Conclusion and Future Work

6.1 Summary

In this project, we explored how data management and machine learning techniques can be applied to analyze large-scale review data and identify suspicious patterns. By combining multiple datasets and applying preprocessing, feature engineering, and anomaly detection, we were able to highlight reviews that deviate from expected behavior.The results showed that suspicious reviews are better understood through a combination of features such as rating deviation, review characteristics, and user activity, rather than any single factor. Overall, this project helped us build a complete data pipeline and apply concepts from the course in a practical setting, while also gaining a deeper understanding of how real-world data challenges are addressed.

6.2 Future Work

There are several directions in which this project can be improved and extended. One potential improvement is the use of labeled datasets, which would allow for more accurate evaluation and the application of supervised learning techniques. Additionally, more advanced natural language processing methods could be used to better capture the complexity of review text, including context and nuanced sentiment.Another area for future work is the detection of coordinated behavior across multiple users, which could help identify groups contributing to suspicious activity. Expanding the dataset and incorporating more features could also improve the robustness and generalizability of the model.